

Program : Diploma in Chemical Engineering	
Course Code : 6071A	Course Title : Plant Utilities
Semester : 6	Credits: 4
Course Category : Program Elective	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- To impart an insight to the common utilities used in process plants.

Course Prerequisites:

Topic	Course code	Course name	Semester
Fluid handling equipments.		Fluid Mechanics	3
Heat transfer equipments		Process Heat Transfer	4
Mass transfer equipments		Mass Transfer Operations I Mass Transfer Operations II	4 5

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the process utilities air and water.	12	Understanding
CO2	Summarize steam system.	18	Understanding
CO3	Summarize the maintenance management of a process plant	18	Understanding
CO4	Summarize plant layout.	12	Understanding

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2						
CO2	2						
CO3	2						
CO4	2						3

3-Strongly Mapped

2-Moderately Mapped

1- Weakly Mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the process utilities air and water.		
M1.01	Summarize the utility – Air in process industries.	3	Understanding
M1.02	Summarize humidity and its application in process plant.	3	Understanding
M1.03	Summarize the utility – Cooling water in process industries	3	Understanding
M1.04	Summarize D M water production.	3	Understanding
Contents: Introduction to utilities - Air - Plant air production –Instrument air-production, Layout – specification – instrument air dryer - buffer vessel – Nitrogen generation system using carbon molecular sieves – Air handling equipments – Compressors, Types – Blower – Fan. Humidity – Absolute humidity, molal humidity, saturation humidity, percentage & relative saturation, adiabatic saturation temp and wet bulb temp - Psychrometric chart- compute humidity, dew point and percentage humidity from humidity chart - Dehumidification Cooling Water – Layout of a cooling tower - Constructional details and working of atmospheric, natural draft, forced draft and induced draft cooling towers - Requirement of blow down and makeup in cooling towers - Cooling water treatment - Chilled Water – Methods of chilled water production, vapour compression system and vapour absorption system. DM water- Illustrate Ion exchange process and membrane process – uses of DM water.			
CO2	Summarize steam system.		
M2.01	Summarize the properties of steam.	2	Understanding

M2.02	Summarize boilers.	8	Understanding
M2.03	Outline steam distribution system.	3	Understanding
M2.04	Summarize boiler feed water system.	4	Understanding
	Series Test – I	1	

Contents:

Properties of steam - Total heat of water - latent heat of water, total heat of steam- super heat - dryness fraction.

Boilers – function - classification of boilers – Illustrate Cochran boiler , Babcock and Wilcox boiler - Illustrate boiler mountings such as steam stop valve, safety valve, water level indicator, pressure gauge and fusible plug - Illustrate accessories such as economizer, feed pump, Super heater, air pre-heater and steam drum – Draft in boilers - natural draft, induced draft, forced draft and balanced draft.

Low pressure, medium pressure and high pressure steam – de super heater – steam traps - types of steam traps - condensate recovery.

Scaling in boilers - Boiler feed water – Deaeration – chemical dosing.

CO3	Summarize the maintenance management of a process plant		
M3.01	Outline the principles of maintenance management.	3	Understanding
M3.02	Summarize the maintenance of major process equipments.	5	Understanding
M3.03	Outline the startup and commissioning of equipments.	5	Understanding
M3.04	Summarize testing methods.	5	Understanding

Contents:

Importance of maintenance management– types of maintenance - breakdown maintenance – scheduled maintenance - preventive maintenance, advantages of preventive maintenance, maintenance schedule – predictive maintenance

Maintenance of major chemical plant equipments – centrifugal pumps, reciprocating pump, shell and tube heat exchanger.

Start up and commissioning of equipments – plant inspection – commissioning of piping and pipeline – commissioning of pressure vessels – commissioning of storage tanks – startup of centrifugal pump, reciprocating pump and shell and tube heat exchanger.

Testing methods– non destructive testing methods, visual testing, leak test, pressure test, liquid penetration test, magnetic particle test, radio graphic test, and ultrasonic test.

CO4	Summarize plant layout.		
M 4.01	Summarise the selection of plant location.	2	Understanding

M 4.02	Summarise Environmental Impact assessment.	3	Understanding
M 4.03	Outline master plot plan and unit plot plan.	3	Understanding
M 4.04	Outline offsite operations.	3	Understanding
	Series Test – II	1	
Contents: Location of Chemical Plant - Geographical factors – Distance from establishments. Environmental Impact Assessment (EIA). Factors to be considered in the preparations of a plant lay out - List the safety factors involved in plant layout - Illustrate master plot plan and unit plot plan – hazardous area classification – electrical area classification. Summarise storage of liquids - open storage tanks - atmospheric storage tanks, fixed roof storage tanks, internal floating roof storage tanks, and external floating roof tank – Summarise pressurized storage for liquids and gases - storage bullets -spheres. Group activity (Identify and prepare the plant layout of a nearby chemical industry and assess the safety factors involved in that plant layout)			

Text / Reference:

T/R	Book Title/Author
T ₁	General Engineering; Prof.Benchamin
R ₁	Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero

